

Policy update on a **fail-only** trajectory: Vanilla GRPO vs SeRPO

Same failed rollout (final outcome = fail). Each column walks reward → advantage $\hat{A}_i$ → PPO effect on π .

shared trajectory: s_1 explore · s_2 fetch · s_3 premature action (fails) · s_4 retry

✗ Vanilla GRPO (outcome reward)

1 · REWARD — ONE SCALAR FOR THE WHOLE TRAJECTORY

$r = 0$

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↓ group-normalize: $\hat{A} = (r - \text{mean})/\text{std}$

2 · ADVANTAGE $\hat{A}_i$ — ALL ROLLOUTS FAILED → STD = 0

0

0

0

0

↓ PPO: $\nabla \propto \hat{A} \cdot \nabla \log \pi$

3 · EFFECT ON POLICY π

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$\hat{A} = 0$ on every token → gradient = 0 → **no parameter update**. The failing step s_3 is never down-weighted; the model repeats the same mistake next time.

π unchanged — nothing learned

✓ SeRPO (segment-level reward)

1 · REWARD — ONE PER SEGMENT

5

4

1

4

↓ normalize within trajectory: $\hat{A}$ varies

2 · ADVANTAGE $\hat{A}_i$ — DIFFERS ACROSS SEGMENTS

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↓ PPO: $\uparrow \pi$ if $\hat{A} > 0$, $\downarrow \pi$ if $\hat{A} < 0$

3 · EFFECT ON POLICY π

$\pi \uparrow$

$\pi \uparrow$

$\pi \downarrow$

$\pi \uparrow$

$\hat{A} \neq 0$ within the trajectory → gradient $\neq 0$. PPO **raises** π for the segments that worked and **lowers** π for the failing step s_3 — a targeted update even though the trajectory failed.

π updated — failure step s_3 down-weighted